FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Advanced)-2020

PAPER –1

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

SECTION – A

1. BCD

Sol. In a frame moving with a velocity of 3 m/s along the z–axis, the displacement of the particle from the equilibrium position is given by

$$\Delta\vec{r} = \left(2\hat{i} - \frac{3}{2}\hat{j}\right)\cos 2\omega t\,.$$

2. AD

Sol. Centre of mass should be at the same height in both cases

3. ABCD

Sol. $y = 2A\sin\left(kx - \omega t\right)$

$$\left(\frac{\partial K}{\partial x}\right)_{antinodes} = 2\mu A^2\omega^2\cos^2\omega t$$

$$\left(\frac{\partial U}{\partial x}\right)_{nodes} = 2\mu A^2\omega^2\cos^2\omega t$$

$$\left(\frac{\partial U}{\partial x}\right)_{antinodes} + \left(\frac{\partial U}{\partial x}\right)_{nodes} = 2\mu A^2\omega^2\text{constant}$$

For a point midway between a node and an adjacent antinode

$$\frac{\partial K}{\partial x} = \mu A^2\omega^2\sin^2\omega\text{t}$$

$$\frac{\partial U}{\partial x} = \mu A^2\omega^2\cos^2\omega t$$

Let $y = A\sin(kx-\omega t) + 2A\sin kx\cos\omega t$

$$\frac{\partial K}{\partial x} = \frac{1}{2}\mu\left(\frac{\partial y}{\partial t}\right)^2 = \frac{1}{2}\mu[-2A\omega\sin kx\sin\omega t - A\omega\cos(kx-\omega t)]^2$$

$$\frac{\partial U}{\partial x} = \frac{1}{2}T\left(\frac{\partial y}{\partial t}\right)^2 = \frac{1}{2}V^2\mu[2A\omega\cos kx\cos\omega t + Ak\cos(kx-\omega t)]^2$$

4. ABC

5. AB

Sol. $\frac{1}{F} = \frac{(\mu-1)}{R} = \frac{1}{20}$

$\Rightarrow 20 = \frac{R}{(\mu-1)}$.....(1)

If equiconvex lens of $f = 20$

Then $\frac{1}{20} = (\mu-1)\frac{2}{R}$

From equation (1) and (2) we conclude

$\therefore\ R = 40(\mu-1)$

That option (A) is correct.

$-\frac{1}{f_L} = (\mu-1)\left[\frac{1}{R} - \frac{1}{\infty}\right] = \frac{(\mu-1)}{R}$ for refraction at the convex surface

$\therefore power\ P_L = \frac{1}{f_L} = -\frac{(\mu-1)}{R}$

For reflection at the silvered plane surface

$F_M = \alpha$

$\therefore power\ P_M = 0$

For refraction at the convex surface again

$P_L = \frac{(\mu-1)}{R}$

Hence power of the system

$P = P_L + P_M + P_L$

$= 2P_L + P_M = -2\frac{(\mu-1)}{R}$

$\therefore$ focal length of the system

$F = \frac{1}{P} = -\frac{R}{2(\mu-1)} = -10cm$

$u = -15cm$

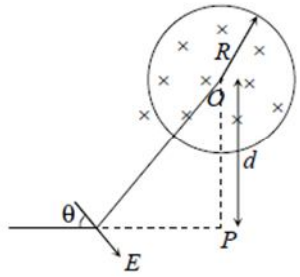
$$\frac{1}{v}+\frac{1}{u}=-\frac{1}{10}$$
$$\frac{1}{v}=\frac{1}{15}-\frac{1}{10}$$
$$\Rightarrow v=-30cm\,so\,B\,is\,correct$$

6. ABC

7. ACD

Sol: $\int \vec{E}.d\vec{l} = A\frac{dB}{dt}$

$$E2\sqrt{x^2+d^2}=\pi R^2 k$$



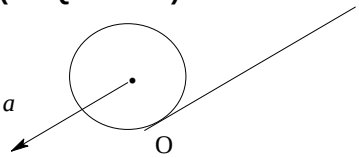
$$E=\frac{\pi R^2 k}{2\sqrt{x^2+d^2}}$$
$$W_{ext}=\int_0^\infty q\vec{E}.dx=\frac{q\pi R^2}{4}k$$

8. C

9. A

10. D

Sol. **(for Q. 8 to 10):**



Taking torques about ICR $\Rightarrow$

$$mg\sin\theta\,.\,R=\left(I_{CM}+mR^2\right)\alpha$$

$$a=\alpha R\Rightarrow\frac{mg\sin\theta\,R^2}{\left(I_{CM}+mR^2\right)}$$

Final $V_{cm}=0\therefore t=\frac{V}{a}=\frac{V\left(I_{CM}+mR^2\right)}{mg\sin\theta R^2}$

Also, Applying energy conservation:-

$$mg\,h=\frac{1}{2}\left(I_{CM}+mR^2\right)\left(\frac{V}{R}\right)^2$$

$$\therefore h=\frac{1}{2}\left[\frac{I_{Cm}}{mR^2}+1\right]\frac{V^2}{g}$$

11. A

12. C

13. C

Sol. **(for Q. 11-13):** The equation of the process BC is

$TV^2 = \text{constant}$

$PV^3 = \text{constant}$

Molar heat capacity of the gas in the process BC is

$$\therefore C=C_V+\frac{R}{1-x}$$

$$=\frac{3R}{2}-\frac{R}{2}=R$$

The work done by the gas during the process BC is

$$\therefore \Delta W_{BC}=\frac{nR\Delta T}{1-x}=\frac{2R(-900)}{1-3}=900R$$

$$\Delta Q_{AB}=nC_p\Delta T=2\times\frac{5R}{2}(1200\text{-}300)=4500R$$

$$\Delta Q_{BC}=nC\Delta T=2\times R(300\text{-}1200)=\text{-}1800R$$

$$\Delta Q_{CA}=\Delta W_{CA}=nRT_0\ln\left(\frac{V_0}{8V_0}\right)=2R300(-3\ln 2)=\text{-}1800R\times 0.7=\text{-}1260R$$

$$\therefore \Delta W_{cycle}=\Delta Q_{cycle}=\Delta Q_{AB}+\Delta Q_{BC}+\Delta Q_{CA}=4500R\text{-}1800R\text{-}1260R=1440R$$

$$\therefore \text{Efficiency},\eta=\frac{\Delta W_{cycle}}{\Delta Q_{supplied}}\times 10=\frac{1440R}{4500R}\times 100=32\%$$

SECTION – B

14. 6

Sol. $$\left(P_0+\frac{4S}{R}\right)\left(\frac{4}{3}\pi R^3\right)=\left(\frac{P_0}{2}+\frac{4S\times 4}{5R}\right)\times\frac{4}{3}\pi\left(\frac{5R}{4}\right)^3$$

$$\Rightarrow P_0+\frac{4S}{R}=\frac{125P_0}{128}+\frac{25S}{4R}\quad\Rightarrow P_0=\frac{96S}{R}=\frac{16nS}{R}$$

$$\therefore n=6$$

15. 8

Sol. $W = U_f - U_i = \frac{Q_0^2}{2C'} - \frac{Q_0^2}{2C}, \qquad v \propto \frac{1}{\sqrt{C}} \Rightarrow C' = \frac{C}{9}$

$= \frac{9Q_0^2}{2C} - \frac{Q_0^2}{2C} = 9E - E = 8E$

$\therefore n = 8$

16. 1

Sol. As impulse exerted by wall is equal to change in momentum of the system.

$F_{av}.\Delta t = 3m.V_{cm} = 3m.\frac{x}{3}\sqrt{\frac{k}{m}}$

$\Rightarrow F_{av} = \frac{x}{\Delta t}\sqrt{km} = \frac{x}{\Delta t}\sqrt{km}.n(given)$

$\therefore n = 1$

17. 2

Sol. $T = \int_0^x \frac{\lambda_0 x}{L} g_{eff}\, dx = \frac{\lambda_0}{2L} x^2 g_{eff}$

$\therefore v = \sqrt{\frac{\lambda_0 x^2 g_{eff}.L}{2L.\lambda_0 x}} = \sqrt{15x}$

$\therefore \int_0^{15} \frac{dx}{\sqrt{15x}} = \int_0^t dt$

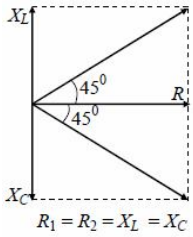
$\Rightarrow t = 2s$

18. 5

Sol. Peak current through $R_1, I_1 = \frac{50}{10\sqrt{2}} = \frac{5}{\sqrt{2}} A$

Peak current through R_2

$I_2 = \frac{50}{10\sqrt{2}} = \frac{5}{\sqrt{2}} A$



Phase difference between $I_1 \text{ and } I_2$ is $\frac{\pi}{2}$

$\therefore$ Peak current through the source is

$I = \sqrt{I_1^2 + I_2^2} = \sqrt{\left(\frac{5}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = \frac{5}{\sqrt{2}} \times \sqrt{2} \quad \therefore \quad I = 5A$

Chemistry

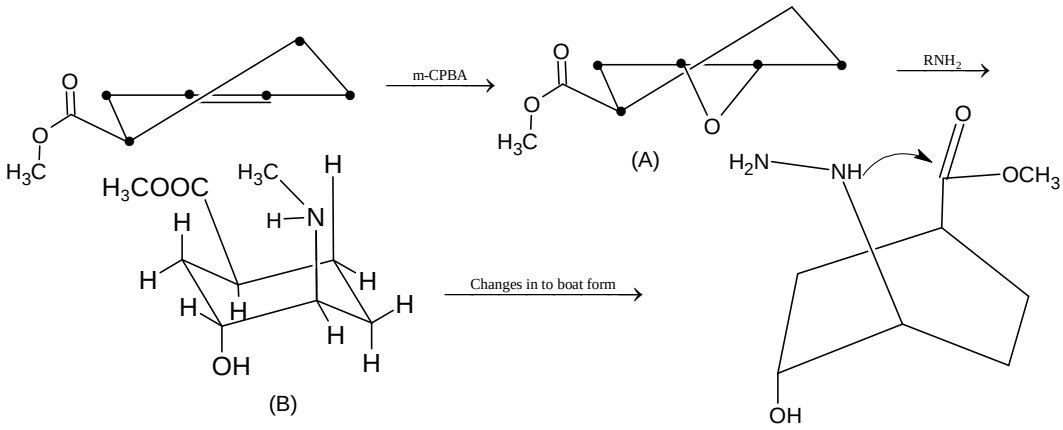
PART – II

SECTION – A

19. AB

I>II , because strain is associated with nitrogen, III>IV because III is less stable as lone pair of nitrogen cannot show the resonance with antibonding MO of C=O

20. ABCD



21. BC

Sol. $Z=\frac{V_{m,real}}{V_{m,ideal}}$

at high temperature and low pressure the gas follows PV = nRT

22. AB

Sol. $Zn+H_2SO_4 \longrightarrow ZnSO_4+H_2$

$2Na+H_2SO_4 \longrightarrow Na_2SO_4+H_2$

$2Al+6H_2SO_4 \longrightarrow Al_2(SO_4)_3+3SO_2+6H_2O$

23. ABC

Sol. 0.005mole NaOH$+0.00235$mole $Na_2CO_3(0.25/106)$

so H+ required = 0.0073 mole

$0.00735= x\times15\times10^{-3}$

$x=0.49$

0.5gm (0.0059 mol)$NaHCO_3$ is added. so total mole of $NaHCO_3$ =0.00825

$0.00825= y\times25.5\times10^{-3}$

$y=0.323$

$\frac{x}{y}=1.5$

$0.00736=0.32\times V(L)$

$y=23ml$

$0.03211=0.49\times V\times10^{-3}$

$V=65.5ml$

24. A

Sol. Volume of H_2SO_4 used at equivalence point is 15mL.

$15\times 0.2\times 2 = 20\times conc.$

Conc. = 0.3

25. C

Sol. $NH_3 \uparrow \rightleftharpoons \frac{1}{2}N_2 \uparrow + \frac{3}{2}H_2 \uparrow$

$n \qquad 0 \qquad 0$

$n(1-\alpha) \quad \frac{n\alpha}{2} \quad \frac{3n\alpha}{2}$

Partial pressure at equation $\left(\frac{1-\alpha}{1+\alpha}\right)P; \frac{\alpha}{2(1+\alpha)}P; \frac{3}{2}\left(\frac{\alpha}{1+\alpha}\right)P$

$$K_p = \frac{\left(P_{N_2}\right)^{1/2}\left(P_{H_2}\right)^{3/2}}{P_{NH_3}} = \frac{3\sqrt{3}P}{4}\left[\frac{\alpha^2}{1-\alpha^2}\right]$$

$$\therefore \alpha = \left[1+\frac{3\sqrt{3}}{4}\frac{P}{K_p}\right]^{-1/2}$$

26. B

27. C

28. C

Sol. **(for Q. 26 - 28):**

$\text{Dolomite} : CaCO_3 .MgCO_3,\ CaCO_3 \xrightarrow{\Delta} CaO + CO_2$

$\text{Cassiterite} : SnO_2 + C \xrightarrow{\Delta} SnO + CO$

$\text{Chalcocite} : Cu_2S + 2Cu_2O \xrightarrow{\text{self reduction}} 6Cu + SO_2$

29. C

30. B

31. C

Sol. **(for Q. 29 – 31):**

(I) Isothermal expansion

(II) Water & Ice will remain in equilibrium to 0°C & 1 atm pressure so $\Delta G = 0$

(III) For adiabatic reversible $T_2P_2^{\frac{1-\gamma}{\gamma}} = T_1P_1^{\frac{1-\gamma}{\gamma}} \Rightarrow T_2 = 300\left(\frac{P_1}{P_2}\right)^{-2/5} = 300\left(\frac{10}{1}\right)^{-2/5} = 120K$

For adiabatic irreversible

$$C_v(T_2-T_1) = -P_{ext}(V_2-V_1) = -P_2\left[\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right]$$

$$\frac{C_v}{nR}(T_2-T_1) = -P_2\left[\frac{T_2}{P_2}-\frac{T_1}{P_1}\right]$$

$$\frac{(T_2-T_1)}{\gamma-1} = -T_2+\frac{T_1P_2}{P_1}$$

$$T_2 = T_1\left[\frac{(\gamma-1)}{\gamma}\frac{P_2}{P_1}+\frac{1}{\gamma}\right]$$

$T_2 = 192K$

(IV) Mixing of ideal gas at constant T & P in an isolated container

$\Rightarrow q = 0, w = 0, \Delta U = 0, \Delta S_{system} > 0, \Delta G < 0$

SECTION – B

32. 4

Sol. ii, iii, v and vi

33. 4

Sol. ii, iii, v, vi are enantiomers

34. 3

Sol. (a) true, (b) true, (c) false, (d) false, (e) false, (f) true

For d : G_{molar} of solvent molecules in solution = G_{molar} of solvent molecules in vapour $\neq$ G_{molar} of solution.

35. 2

Sol. $$\mu = \frac{m_{e^-} \times m_{e^+}}{m_{e^-} + m_{e^+}} = \frac{m_e}{2}$$

$$IE = \frac{\mu \times 13.6}{m_e}\text{eV} = \frac{\frac{m_e}{2} \times 13.6}{m_e} = 6.8\text{eV}$$

$A = 6.8$

36. 5

Sol. iii , iv, v, vii is correct. P4O10 is a dehydrating agent.

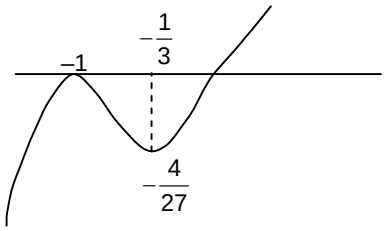
Mathematics

PART – III

SECTION – A

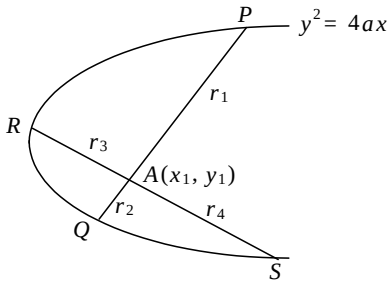
37. AC

Sol.



38. ABD

Sol. Let two perpendicular chords through $A(x_1, y_1)$ be PQ and RS.



Equation of PQ is $\frac{x-x_1}{\cos\theta} = \frac{y-y_1}{\sin\theta} = r$ where $\tan\theta$ = slope of PQ.

Any point on this line may be taken as $(x_1 + r\cos\theta, y_1 + r\sin\theta)$. As the point lies on $y^2 = 4ax$,

$\therefore \quad (y_1 + r\sin\theta)^2 = 4a(x_1 + r\cos\theta)$.

$\Rightarrow \quad r^2\sin^2\theta + 2y_1\sin\theta.r - 4a\cos\theta r + y_1^2 - 4ax_1 = 0$

$\therefore \quad r_1r_2 = \frac{-(y_1^2 - 4ax_1)}{\sin^2\theta}$

Similarly $r_3r_4 = \frac{-(y_1^2 - 4ax_1)}{\sin^2\alpha}$ where $\tan\alpha$ = slope of RS

Since PQ is perpendicular to RS,

$\therefore \quad \alpha = 90 + \theta$ or $\theta - 90$

In either case; $\sin^2\alpha = \cos^2\theta$

Now $\frac{1}{r_1r_2} + \frac{1}{r_3r_4} = \frac{\sin^2\theta + \cos^2\theta}{-(y_1^2 - 4ax_1)} = \frac{1}{-(y_1^2 - 4ax_1)}$ which is constant

39. BD

Sol. Curve is $(y-2)^2 = 2x^3 - 4$ (1)

$$\frac{dy}{dx} = \frac{3x^2}{y-2}$$

$\therefore$ Equation of tangent at (x_1, y_1)

$$y - y_1 = \frac{3x_1^2}{y_1 - 2}(x - x_1)$$

As it passes through (1, 2)

$\Rightarrow \quad 2 - y_1 = \frac{3x_1^2}{y_1 - 2}(1 - x_1)$

or $\quad (y_1 - 2)^2 = 3x_1^2(x_1 - 1)$

or $\quad 2x_1^3 - 4 = 3x_1^3 - 3x_1^2$ [As (x_1, y_1) lies on the curve]

or $\quad x_1^3 - 3x_1^2 + 4 = 0$

or $(x_1 + 1)(x_1 - 2)^2 = 0$

$\therefore$ $x_1 = -1$ or 2

But $x_1 \neq = -1$ (it is clear from equation (1))

$\therefore$ $x_1 = 2$

$\therefore$ $y_1 = 2 \pm 2\sqrt{3}$

Points of contact are $(2, 2 + 2\sqrt{3})$ and $(2, 2 - 2\sqrt{3})$

$\therefore$ sum of x–coordinates of points of contact is 4

$\therefore$ sum of y–coordinates of points of contact is 4.

40. BC

Sol. If $\left|\int_a^b f(x)dx\right| = \int_a^b |f(x)|dx$

$\Rightarrow$ $f(x)$ is either positive in (a, b)

or $f(x)$ is negative in (a, b) and also $b > a$

Also $g'(x) = f(x)$

When $f(x) > 0 \Rightarrow g'(x) > 0 \Rightarrow g(x)$ is increasing function

When $f(x) < 0 \Rightarrow g'(x) < 0 \Rightarrow g(x)$ is decreasing function

also $g(0) = 0$

$$\int_a^b f(x)g(x)dx = \int_a^b g(x)g'(x)dx$$

$$= \frac{(g(b))^2 - (g(a))^2}{2}$$ which will be positive for $0 < a < b$.

41. AC

Sol. $C_1 \to C_1 - C_2$

$$\Rightarrow \Delta = \begin{vmatrix} 4 & (e^{i\alpha} - e^{-i\alpha})^2 & 4 \\ 4 & (e^{i\beta} - e^{-i\beta})^2 & 4 \\ 4 & (e^{i\gamma} - e^{-i\gamma})^2 & 4 \end{vmatrix} = 0$$ ($\because$ C_1 & C_3 are identical).

42. AC

Sol. We have AB = A

$\Rightarrow$ A (BA) = A [$\because BA = B$]

$\Rightarrow$ (AB) A = A

$\Rightarrow$ AA = A $[\because AB = A]$

$\Rightarrow$ $A^2 = A$

Again BA = B

$\Rightarrow$ B (AB) = B $[\because AB = A]$

$\Rightarrow$ (BA) B = B

$\Rightarrow$ BB = B $\Rightarrow$ $B^2 = B$

43. BD

Sol. $|z| - \log_2(z + 2) = 0$

$z + 2 = 2^{|z|}$

$x + 2 = 2^x$ and $y = 0$

hence $x = 2$, $y = 0$.

44. D

45. A

46. C

Sol. **(for Q. 44 to 46):**

(I) $\left(2+\frac{1}{2}+\frac{1}{2^3}+.....\right)+\left(\frac{1}{2^2}+\frac{1}{2^4}+....\right)$

(II) $f(t)=f^{-1}(t)$ $(t=\sin x)$ (t=1)

(III) $x=2n\pi+\frac{\pi}{4}$

(IV) $\alpha+\beta=a-1+\frac{4}{a-1}+1\geq 5$

(i) $\sum_{n=0}^{\infty}\tan^{-1}\sqrt{n+2}-\tan^{-1}\sqrt{n}$

(ii) $a^2+b^2+c^2<2(ab+bc+ca)$

(iii) $(ab+bc+ca)(a-b)=0$

$\Rightarrow ab+bc+ca=0$

(iv) $P^2+Q^2+R^2+S^2=1$

47. D

48. A

49. C

Sol. **(for Q. 47 to 49):**

(I) $|x|+|2y|=2$ and points are $\left(\pm\frac{2}{5},\pm\frac{4}{5}\right)$

(II) $(x-3)^2+(y-4)^2=4$ required minimum and maximum slopes of target from origin

(III) $3[(x-1)^2+(y-2)^2+10]$

(IV) Points are on

$|Z|=\frac{2}{\sqrt{3}}$

$AP^2+BP^2+CP^2=3|Z|^2+|Z_1|^2+|Z_2|^2+|Z_3|^2$

$-Z(\bar{Z}_1+\bar{Z}_2+\bar{Z}_3)-Z(Z_1+Z_2+Z_3)$

$=1+3\times\frac{4}{3}=5$

(i) Minimum distance

$\sqrt{4+(\tan\alpha-\cot\alpha)^2}-\sqrt{3}=2-\sqrt{3}$

(ii) $n_2=1-\frac{1}{1+n_1}$

(0 ,0) and (-2, 2)

(iii) $T_n\ \frac{\sqrt{n^2+(n-1)^2}-\sqrt{n^2-(n-1)^2}}{4}$

(iv) $(x+y-5)(3x+2y)(3x-2y)=0$

SECTION – B

50. 2

Sol. $\tan^{-1}x+\tan^{-1}\frac{1}{y}=\sin^{-1}\frac{3}{\sqrt{10}}$

$\tan^{-1}\frac{1}{y}=\tan^{-1}3-\tan^{-1}x$

$\tan^{-1}\frac{1}{y}=\tan^{-1}\frac{3-x}{1+3x}$

$y=\frac{3x+1}{3-x}$ x=1,2 and corresponding y=2,7, thus two solution

51. 3

52. 3

Sol. $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}$

$$A^2=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}=\begin{bmatrix}9 & 8 & 8\\ 8 & 9 & 8\\ 8 & 8 & 9\end{bmatrix}$$

$$4A=\begin{bmatrix}4 & 8 & 8\\ 8 & 4 & 8\\ 8 & 8 & 4\end{bmatrix}$$

$$A^2-4A-5I=0$$

$$\begin{bmatrix}9 & 8 & 8\\ 8 & 9 & 8\\ 8 & 8 & 9\end{bmatrix}-\begin{bmatrix}4 & 8 & 8\\ 8 & 4 & 8\\ 8 & 8 & 4\end{bmatrix}-\begin{bmatrix}5 & 0 & 0\\ 0 & 5 & 0\\ 0 & 0 & 5\end{bmatrix}=\begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}=0 \qquad A^2-4A-5I=0$$

Multiplying by A^{-1} both sides,

$$A^{-1}\left[A^2-4A-5I\right]=A^{-1}0$$

$$A-4I-5A^{-1}=0$$

$$A-4I=5A^{-1}$$

$$A^{-1}=\frac{1}{5}(A-4I)=\frac{1}{5}\left(\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{bmatrix}-\begin{bmatrix}4 & 0 & 0\\ 0 & 4 & 0\\ 0 & 0 & 4\end{bmatrix}\right)$$

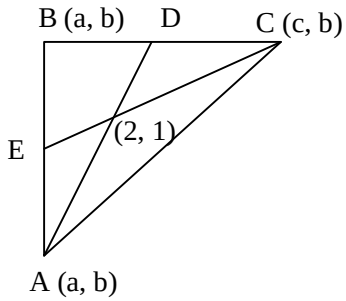
$$A^{-1}=\frac{1}{5}\begin{bmatrix}-3 & 2 & 2\\ 2 & -3 & 2\\ 2 & 2 & -3\end{bmatrix}.$$

53. 7

Sol. Let equation of new plane $2x-2y+z-3+\lambda z=0$

Point (3, 1, 1) lie on $it\Rightarrow\lambda=-2$

Hence equation of new plane 2x – 2y –z = 3



$$\cos\alpha=\frac{4+4-1}{3.3}=\frac{7}{9}$$

54. 4